

# **Long Thanh International Airport Passenger Terminal Construction Project**

## **Interface Control Document (ICD) Security Access Control System (SACS)**

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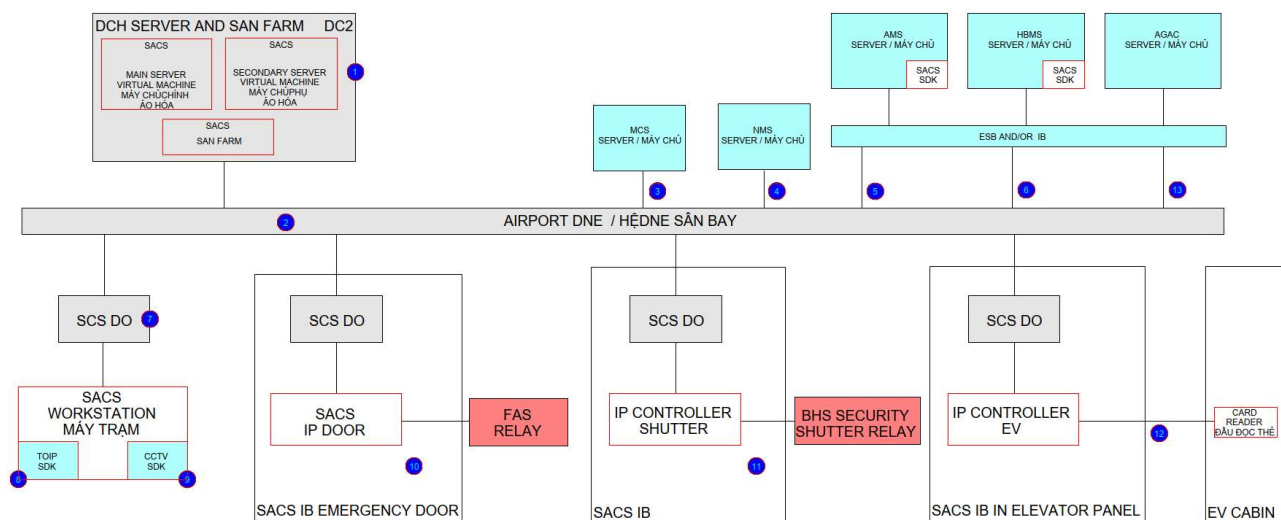
## I. INTRODUCTION

The purpose of this document – Interface Control Document (ICD) – is to control and maintain a record of all interface information generated for the project - Passenger Terminal Construction Project for Long Thanh International Airport.

The aim of this document includes defining and recording all relevant inputs to and all relevant outputs from any of the relevant systems (subsystems and third-party systems) for some potential or actual user of these scope covered systems (and subsystems).

## II. SCOPE OF WORK

### 1. Detail SACS interface diagram



### 2. The following table depicts the scope between SACS Contractor and ICT Contractor

| Description                                                  | ICT Contractor | SACS Contractor |
|--------------------------------------------------------------|----------------|-----------------|
| Location : Datacenter / Server room                          | X              |                 |
| Location : IT Field cabinets (Field Distribution Rooms(FDR)) | X              |                 |
| Location : Physical security                                 | X              |                 |
| Hardware : Servers (VMware environment)                      | X              |                 |
| Hardware : Backup Server                                     | X              |                 |
| Hardware : SAN Storage                                       | X              |                 |
| Hardware : SACS Clients (Control room, Admin PC)             |                | X               |
| Hardware : Racks (Including KVM/UPS)                         | X              |                 |

|                                                                                                               |   |   |
|---------------------------------------------------------------------------------------------------------------|---|---|
| Hardware : Network Active Equipment (Firewall/Core/Access switches)                                           | x |   |
| Hardware : Network Passive Equipment (Horizontal cabling, UTP)                                                | x |   |
| Hardware : Network Passive Equipment (Vertical cabling, FO)                                                   | x |   |
| Hardware : Network Wifi                                                                                       | x |   |
| Hardware : Videowall monitor                                                                                  | x |   |
| Hardware : SACS Controller, Card reader, Biometric reader, Card printer, Local Alarm, IB, Card, Duress button |   | x |
| Software : VMware Hypervisor and Licenses                                                                     | x |   |
| Software : SACS Main Software and Backup Software licences                                                    |   | x |
| Software : SACS Client Licenses                                                                               |   | x |
| Software : SACS HA Nerverfail Licenses                                                                        |   | x |
| Software : Windows licenses                                                                                   | x |   |
| Software : SQL Server License                                                                                 | x |   |
| Security : Anti virus/ malware software and licenses                                                          | x |   |
| Security : Network design (VLAN'S)                                                                            | x |   |

### III. SYSTEM TO BE INTERFACED WITH SACS

Control the elevator (EV)

Fire Alarm System (FAS)

Structured Cabling Network (SCN)

Structured Cabling System (SCS)

Data Communication Hardware (DCH)

Enterprise Service Bus (ESB)

Information Broker (IB)

Master Clock System (MCS)

Simple Network Management Protocol (SNMP)

Airport Management System (AMS)

Data Network Equipment (DNE)

High Level Design for Building Management System (HBMS)

Telephone over Internet Protocol (ToIP)

Baggage Handling System (BHS)

Enterprise Resource Planning (ERP)

Closed Circuit TeleVision System (CCTV)

Automatic Gate Access Control (AGAC)

Public Address System (PAS)

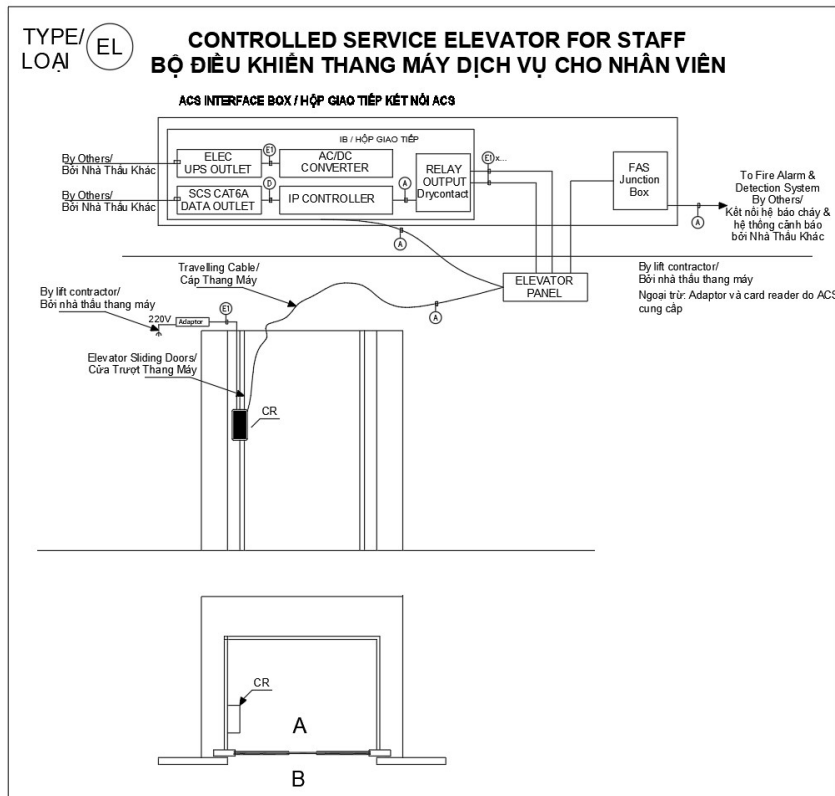
Automated Boarding Area Access Gate (ABAAG)

Automated Border Control Gate (ABCG)

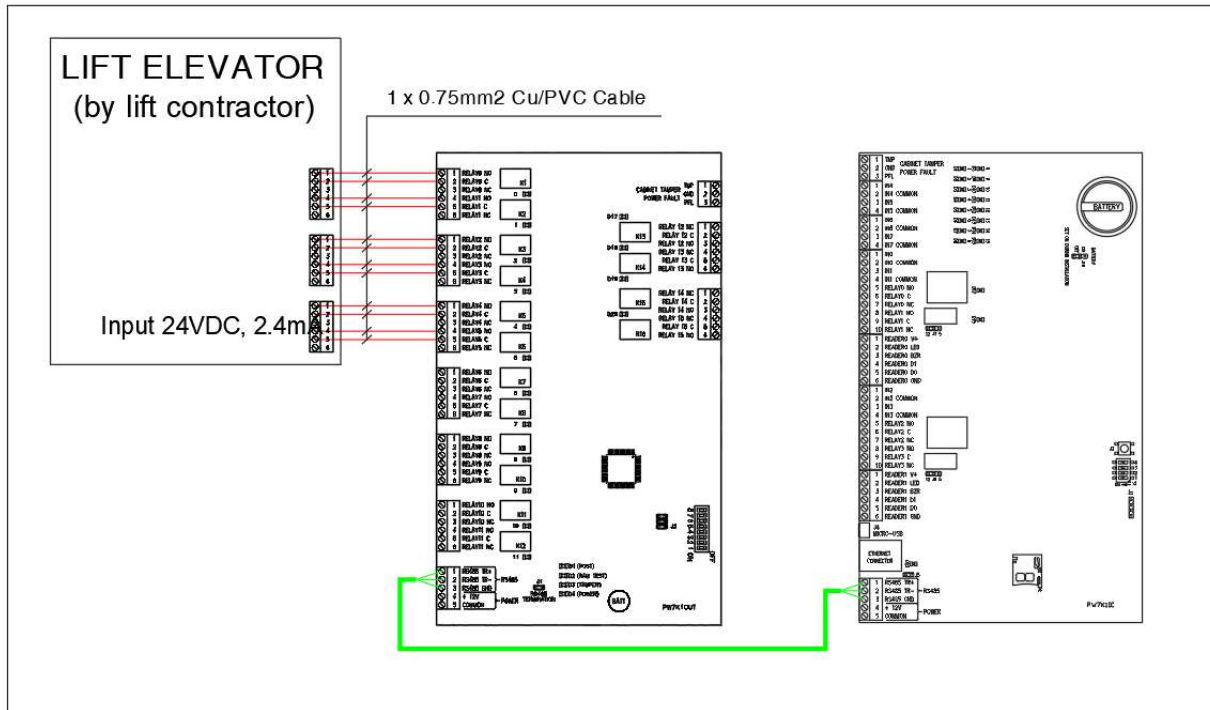
Automated Self-Boarding Gate (ASBG)

Boarding and Deplaning Route Management (BDRM)

## III.1. Control the elevator (EV)



|   |                                                                                                       |
|---|-------------------------------------------------------------------------------------------------------|
| Ⓐ | 1 x 1 pair 18 AWG cable<br>1 x Cáp tín hiệu 1 pair 18AWG                                              |
| Ⓔ | 2 x 1.5mm <sup>2</sup> Cu/PVC<br>2 x 1.5mm <sup>2</sup> Cu/PVC<br>x... Số lượng tín hiệu theo số tầng |
| Ⓔ | 2 x 2.5mm <sup>2</sup> Cu/PVC<br>2 x 2.5mm <sup>2</sup> Cu/PVC                                        |
| Ⓓ | 1 x CAT 6 UTP<br>1 x CAT 6 UTP                                                                        |

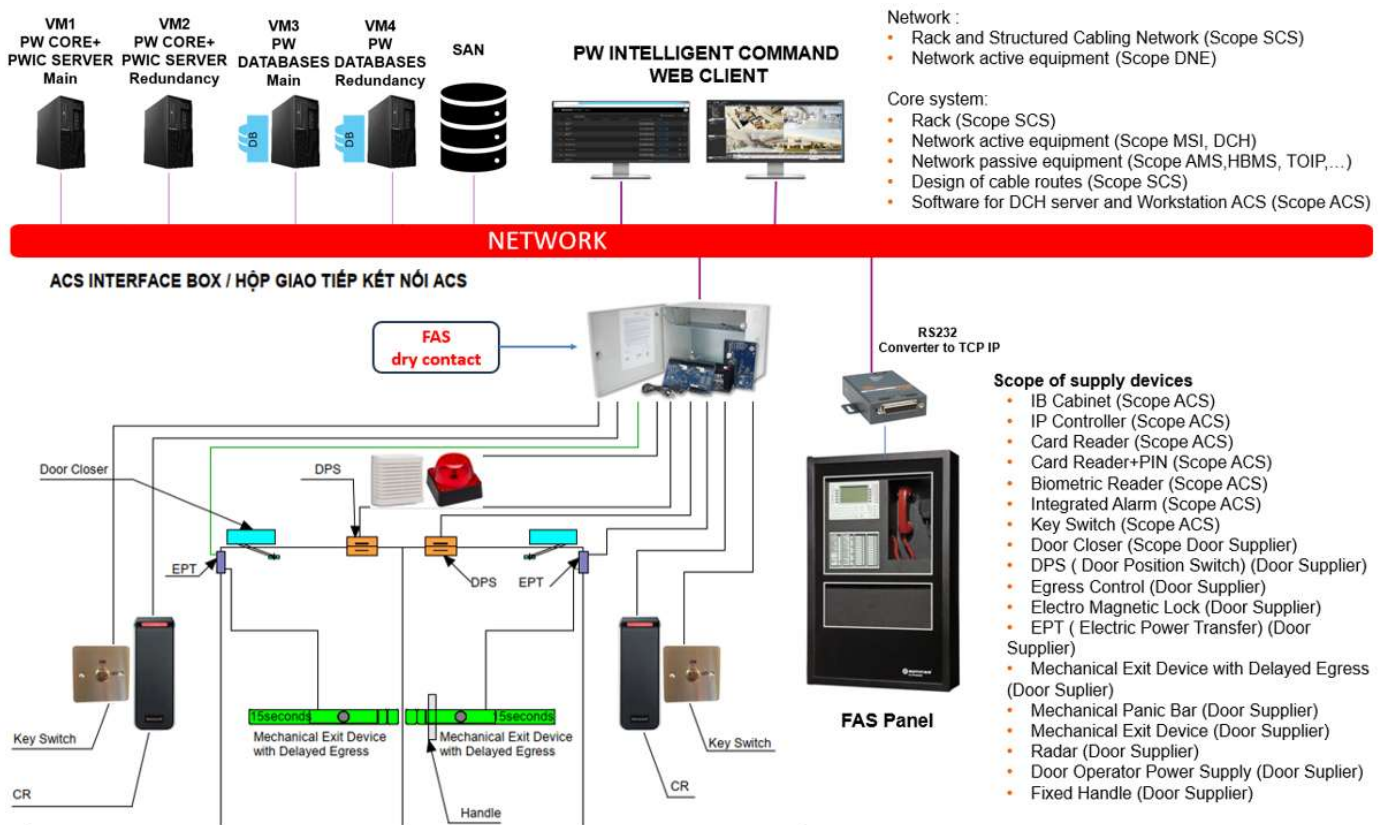


The SACS system will control the elevator system through dry contact.

## III.2. Fire Alarm System (FAS)

The SACS system will connect to the FAS system through two protocols.

- (Through A Software interface the FAS shall report the fire zones in alarm status to the ACS). Pro-Watch becomes the central integration engine (not the HSDK middleware). Each Fire Panel is now able to independently report and integrate to Pro-Watch. It is not reliant on 4 pieces of software in between. Generic Channel approach is much more direct. Please see the line diagram that explains.
- The FAS system will provide for each emergency door connected to ACS a dry contact to release the door in case of emergency evacuation.



SACS and FAS system connection diagram

## III.3. Structured Cabling Network (SCN) and Structured Cabling System (SCS)

ACS uses transmission cable infrastructure provided by SCN; details of cable types are according to updated drawings.

The ACS will use the cabling provided by the Structural Cabling System (SCS); The ACS system provider will use the SCS infrastructure and will connect its devices to the exit device outlets provided by SCS.



## III.4. Baggage Handling System (BHS)

The BHS system shall provide for each security shutter connected to ACS a dry contact to indicate the status of shutter opening.

Additionally, the ACS system can connect to the BHS system via API or SDK protocol

## III.5. Data Communication Hardware (DCH)

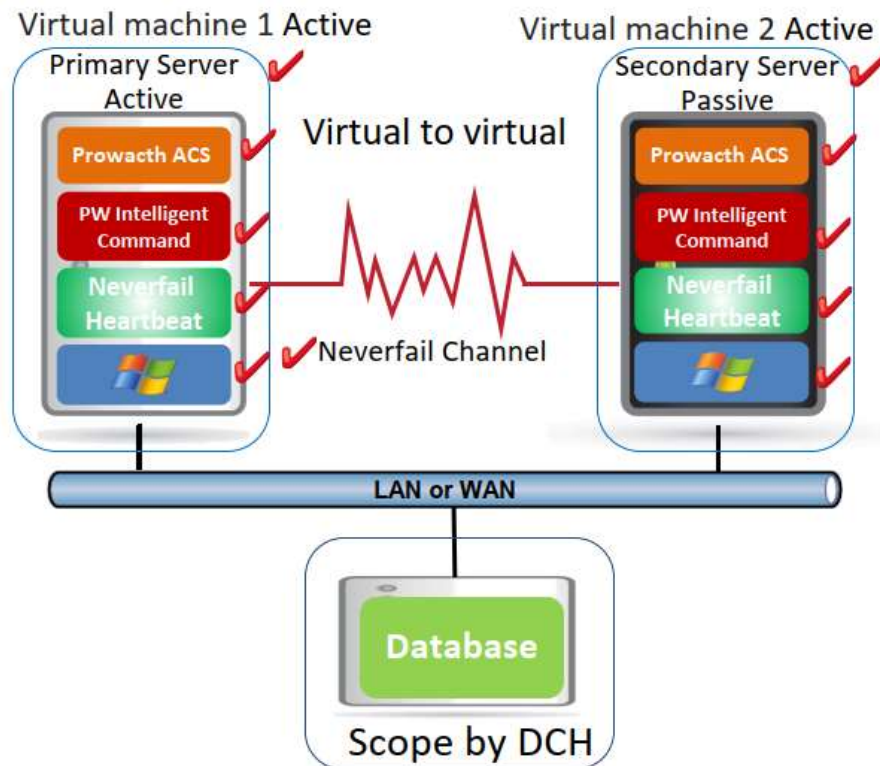
The ACS system shall be virtualised and hosted in the DCH Server Farm;

ACS management system to be virtualized in the DCH Server Farm

The ACS shall use the DCH San Storage Farm for data Storage

The central system shall be installed on the virtualized environment. The ACS coordinates with DCH contractor for hardware requirement subject for virtualization. A high availability server solution shall be proposed as an option.

All the central ACS equipment (Management System) shall be virtualized and supported by DCH Server Farm. The ACS server and storage shall be installed on the DCH server and san farm. The system provider should provide virtualized applications.



Scope/ Phạm vi:

- Virtual machine, operating system, database by DCH/ Máy ảo, hệ điều hành, cơ sở dữ liệu của DCH
- Software core system ACS by SACS/ Hệ thống phần mềm trung tâm ACS của SACS



## 2 x Virtual Machine for PW CORE SERVER + PW IC CORE SERVICES / PW IC APP / PW IC WEB

- CPU: 2 X 2nd generation Intel Xeon scalable processors with silver/gold/platinum (32 cores or greater) 3.6 GHz max turbo frequency
- RAM: 128 GB Recommended
- SSD/HDD: 500GB for OS
- SSD RAID 4TB (For Database)
- Minimum 1Gbps For Networks
- OS: Windows Server 2019/2022 Standard
- SQL Server: 2019/ 2022 Standard

## PW / PW IC CLIENT

- CPU: Intel® Core i3, i5, or i7 Processor 2.8 GHz or higher
- RAM: 32 GB
- Monitor Resolution: 1920x1080 pixels; 24 bit color or higher / Standard VGA Graphics Adapter (Display Adapter)
- OS: Windows 10/11 (32 or 64 bit) / Windows 2016/2019 Server Enterprise or Standard edition (32 or 64 bit)
- Browser: Chrome V80.0.3987.106 or above

**Recommended Specifications**

| Scope of 5.11                                           |       |          |             |              |                                                                     |
|---------------------------------------------------------|-------|----------|-------------|--------------|---------------------------------------------------------------------|
| Recommended Specifications for Virtual Machine DCH 5.11 |       |          |             |              |                                                                     |
| VM                                                      | vCore | Ram (GB) | SSD/VM (GB) | HDD SAN (GB) | OS System                                                           |
| VM 1<br>CORE SERVER +PW IC APP                          | 64    | 128      | 500 GB      |              | Windows Server 2019/2022 standard                                   |
| VM 2<br>CORE SERVER +PW IC APP                          | 64    | 128      | 500 GB      |              | Windows Server 2019/2022 standard                                   |
| VM 3<br>DATABASE SERVER                                 | 32    | 64       | 500 GB      | 4000         | Windows Server 2019/2022 standard<br>SQL Server 2019/ 2022 Standard |
| VM 4<br>DATABASE SERVER                                 | 32    | 64       | 500 GB      |              | Windows Server 2019/2022 standard<br>SQL Server 2019/ 2022 Standard |

HA solution for Core Server + PW IC APP servers within ACS scope

HA solution for Database Server within DCH scope

## III.6. Other Systems

Enterprise Service Bus (ESB)

Information Broker (IB)

Master Clock System (MCS)

Simple Network Management Protocol (SNMP)

Airport Management System (AMS)

Data Network Equipment (DNE)

High Level Design for Building Management System (HBMS)

Telephone over Internet Protocol (ToIP)

Baggage Handling System (BHS)

Enterprise Resource Planning (ERP)

Closed Circuit TeleVision System (CCTV)

Automatic Gate Access Control (AGAC)

Public Address System (PAS)

Automated Boarding Area Access Gate (ABAAG)

Automated Border Control Gate (ABCG)

Automated Self-Boarding Gate (ASBG)

Boarding and Deplaning Route Management (BDRM)

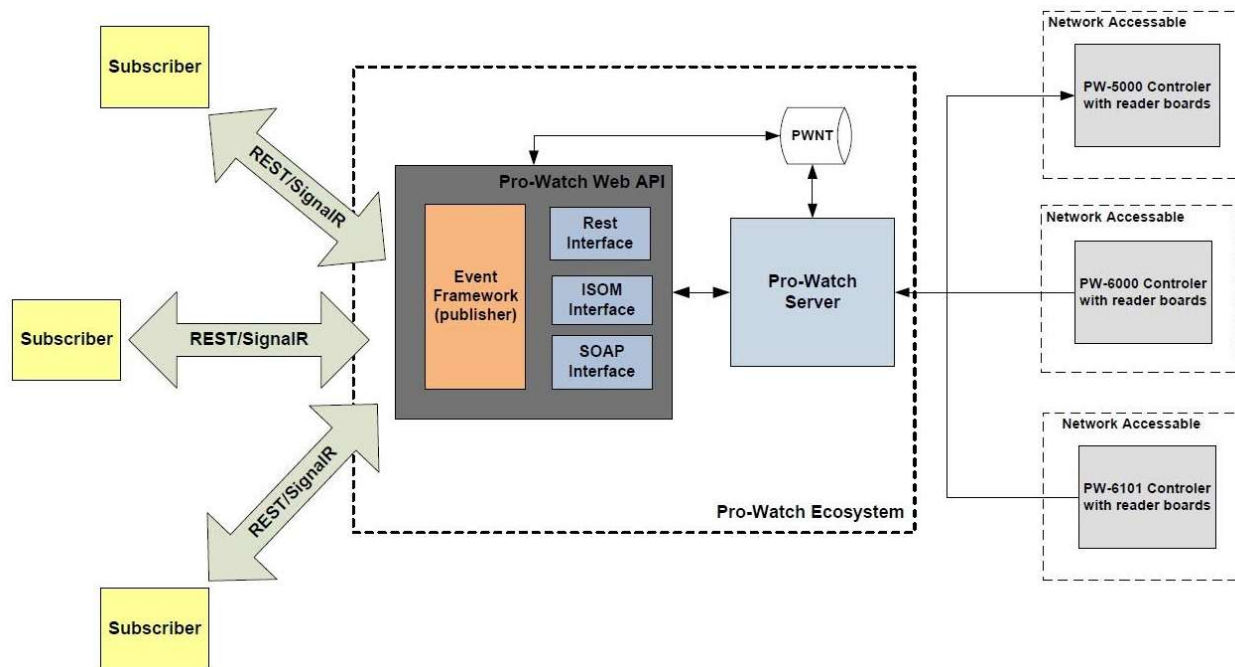
## IV. API INTEGRATION DOCUMENTATION

### 1. General

The purpose of integrating ACS with IB is to centralize access event data, standardize communication, ensure identity and access synchronization, and enable ACS data sharing with other systems (CCTV, Fire Alarm, etc.) through a single integration channel, ensuring scalability, security, and centralized management.

The main objectives of the solution include:

- Seamlessly connect access data and security events between the ACS system and other systems (CCTV, BMS, Fire Alarm, HRM, FRS, SCADA, AODB...) through the IB intermediary system to ensure centralized monitoring and quick response through real-time event transmission, reception, and synchronization, helping the security monitoring, operation, and management system have a comprehensive view.
- Ensure data safety, security, and consistency throughout the entire information exchange process between systems through IB – acting as an intermediary layer to coordinate, control API access, and standardize data.
- Enhanced future scalability and integration, allowing ACS to easily connect to other operational or security systems in the airport infrastructure without changing the existing structure.



## 2. Integration solution between ACS and Integration Broker (IB)

The Access Control System (ACS - Honeywell Pro - Watch) at Long Thanh International Airport is integrated with other systems through Integration Broker (IB), an intermediary platform that allows connection, exchange of data, and events between operational systems in the airport.

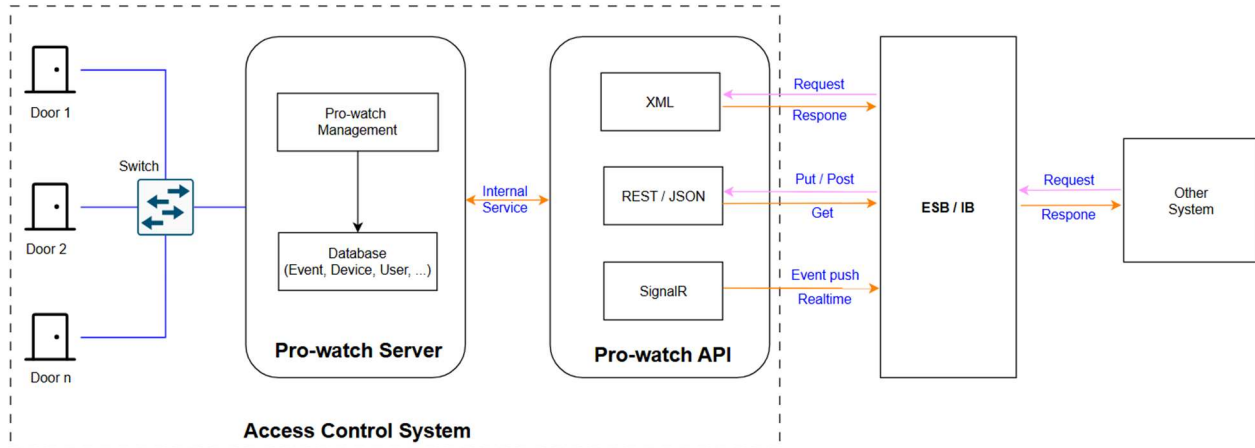
Integration Broker acts as the sole communication intermediary between ACS and other systems, enabling:

- Standardize data formats (JSON, XML).
- API access control and authentication.
- Manage real-time event streams and periodically synchronize data.
- Ensure security and scalability when adding new systems.

## 3. Scope

| Task                                                  | SACS Contractor | IB Contractor |
|-------------------------------------------------------|-----------------|---------------|
| Describe the integration interface between ACS and IB | x               |               |
| Configure ACS hardware (Controller, Reader, etc.)     | x               |               |
| Transmit access events (entry/exit)                   | x               |               |
| Synchronize users                                     |                 | x             |
| Send door control commands                            |                 | x             |
| Integration with other systems                        |                 | x             |

## 4. System diagram



### Access Control System (ACS)

- Main functions: manage users, cards/fingerprints / QR codes, control doors, record entry/exit events.
- Provides APIs (REST / JSON / XML) that allow other systems to access data or issue control commands.
- Serves as the source of access event data (Access Events, Door Status).

### Information Broker (IB)

- The Information Broker (IB) is a middleware layer responsible for collecting, normalizing, and distributing data.
- IB receives data from ACS via API and converts it into a standardized format (common JSON / XML schema).
- IB distributes this data to other systems (CCTV, Fire Alarm, BMS, etc.).
- At the same time, IB transmits commands from other systems to ACS (e.g., emergency door release).

### Other System

- Do not communicate directly with ACS, but only connect through IB.
- Examples: CCTV / VMS: receive events from ACS to trigger camera footage retrieval.
- Each connection will be defined in a separate ICD (Interface Control Document).

## 5. Integration mechanism

ACS provides the Pro-Watch API for IBs to mine data, synchronize events, and trigger security responses.

Main protocols:

- REST / JSON API – serves queries and exchanges data on demand (GET, POST, PUT).
- SignalR Realtime – push events to IB in real time, including event types such as:
  - Door Forced Open (door forced open).
  - Door Held (door open too long).

- Access Denied.
- Access Granted (valid access).
- System Alarm (system alarm).

The Integration Broker (IB) will receive these events, then:

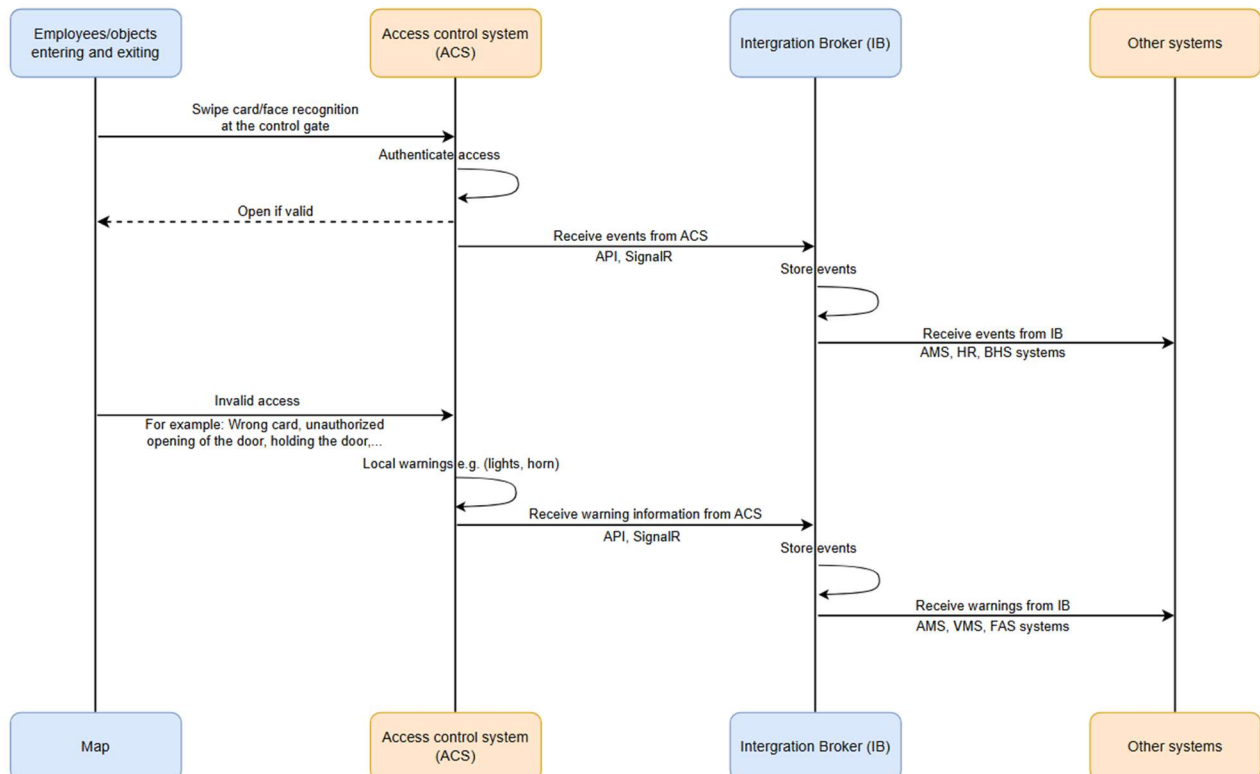
- **Standardization and routing** from ACS and transfer to related systems (CCTV, FAS, BMS, ...).
- **Trigger automated actions** based on event types, for example:
  - ACS <=> CCTV: display video, record 30 seconds before and ≥10 minutes after when there is a violation.
  - ACS <=> FAS: automatically open the exit door when there is a fire alarm.
  - ACS <=> BMS: door status monitoring, device warning.
  - ACS <=> BDRM: manage boarding/disembarking doors, monitor entry/exit events, synchronize CCTV videos.
- **Record events and synchronize data** to the AMS system to serve the general operations of the airport.

## 6. Data processing flow

### 6.1 Model

In an integrated system, data transmission and processing between components play a key role in ensuring synchronization, security, and operational efficiency.

This section describes the process of processing and transmitting event data between the access control system (ACS) and other monitoring and management systems in the airport via the Integration Broker (IB).



## 6.2 Usage flow

### 6.2.1 Employee entry / valid area

| Step | Act                                                               | Detailed description                                                                                                                                                                     | Data transmitted/received                                                                                |
|------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 1    | Staff/person swiping card or using facial recognition at the door | The ACS system collects information about employee code, time, location, action type (in, out), time, ...                                                                                |                                                                                                          |
| 2    | Access authentication                                             | ACS valid authentication: authorizes door opening, records "valid entry/exit" event                                                                                                      |                                                                                                          |
| 3    | Send event data to IB via API or SignalR                          | ACS sends events with information about time, location, employee code, and time to the IB system.                                                                                        | JSON event: employee code, time, location, action type (door location), authentication status, time, ... |
| 4    | IB receives events                                                | IB standardizes data that labels the system identifier, location, time, and event type.                                                                                                  |                                                                                                          |
| 5    | Store and push data to other systems                              | Store events after normalizing data according to the IB system.<br><br>Push events to other systems (AMS, HR, BHS).                                                                      | Event log: employee, time, location, action type (in, out).                                              |
| 6    | Monitoring and verification                                       | AMS: to display, manage door and user status.<br><br>HR: to record attendance history, verify personnel.<br><br>BHS: to synchronize the journey of baggage handlers in restricted areas. | Query related data employee, time, location, type of action (in, out)                                    |

### 6.2.2 Security Breach Detection / Alerting

| Step | Act                                                               | Detailed description                                                                 | Data transmitted/received |
|------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------|
| 1    | The ACS system detects unauthorized access or a security incident | Invalid card swipe, unauthorized opening, after-hours access, suspected intrusion... |                           |

|   |                                                  |                                                                                                                                                                                                          |                                                                                               |
|---|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 2 | Mark events that require a warning               | ACS determines the level, labels the event as a "warning/violation."                                                                                                                                     | Alarm event: alarm type, priority, event code                                                 |
| 3 | Send alert information to IB via API or SignalR. | ACS sends events with information about time, location, employee code, and time to the IB system via API, SignalR                                                                                        | JSON alarm: tag code, time, location, alarm type, risk level, alarm status                    |
| 4 | IB receives events                               | IB standardizes data that labels the system identifier, location, time, and event type.                                                                                                                  |                                                                                               |
| 5 | Store and push data to other systems             | Store warning events after data normalization according to the IB system.<br><br>Push events to other systems (VMS, FAS, AMS).                                                                           | Alarm log: alarm information, status, sent to other systems.                                  |
| 6 | Processing coordination &                        | VMS: receive warnings from IB, display live cameras of related areas, and store events.<br><br>AMS: receives warnings from IB, coordinates flight management operations appropriate to the warning type. | Processing status; processing notes; confirmation log, inspected images/videos (if available) |

### 6.2.3 Action (close/open/lock door)

| Step | Act                                | Detailed description                                                                                         | Data transmitted/received                     |
|------|------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 1    | Other systems send warning events. | Send reverse control commands (close/open door, temporary lock, ...) to IB.                                  | Alarm event: alarm type, priority, event code |
| 2    | IB receives events                 | IB standardizes data that labels the system identifier, location, time, and event type.                      |                                               |
| 3    | Store and send data to ACS         | Store warning events after data normalization according to the IB system.<br><br>Push events to ACS systems. | JSON Push Action: remote door                 |



|   |                           |                                                    |                                                      |
|---|---------------------------|----------------------------------------------------|------------------------------------------------------|
| 4 | Processing coordination & | ACS acts (close, open, hold door temporarily, ...) | Eventlog: information about actions performed on ACS |
|---|---------------------------|----------------------------------------------------|------------------------------------------------------|

#### 6.2.4 Data synchronization and analysis

| Step | Act                                                                | Detailed description                                                                                                                                                                               | Data transmitted/received                                              |
|------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| 1    | ACS continuously records card swipe/entry events at multiple doors | Data includes: employee code, time, location, action type...                                                                                                                                       | Event log: list of events (employee code, time, location, action type) |
| 2    | Pass event data to IB via SignalR, API.                            | ACS valid authentication: authorizes door opening, records “valid entry/exit” event                                                                                                                |                                                                        |
| 3    | Send event data to IB via API or SignalR                           | Periodically or in real time, ACS sends event logs to IB                                                                                                                                           | JSON log                                                               |
| 4    | IB receives events                                                 | IB conducts standardization of system identification labeling data                                                                                                                                 |                                                                        |
| 5    | Storage, synchronization                                           | Store events after normalizing data according to the IB system.<br><br>Synchronize data to the central system.                                                                                     | Mapping: event log                                                     |
| 6    | Retrieval and reporting                                            | Administrators can query and export reports based on criteria (employees, time, area, etc.) with both ACS data and other systems for inspection, analysis, auditing, or investigation when needed. | Reporting data: log summary, metadata, Excel/PDF report files          |

## 7. Functional specifications

### 7.1 List of operations (proposed)

This section describes the main business scenarios during the operation of an access control system (ACS) when it has been integrated with other systems through an Integration Broker (IB).

Each business flow represents how the ACS generates events, the IB processes intermediate data, and other systems respond or take corresponding actions in the actual airport operating environment.

| # | Alert / Event Name                                     | Trigger condition (EventCode) | Minimum ACS data sent via IB                | Related receiving system          |
|---|--------------------------------------------------------|-------------------------------|---------------------------------------------|-----------------------------------|
| 1 | <b>Lost card warning</b>                               | 406 - Lost Card Attempt       | CardID, EventCode, DoorID, Time             | <b>HR, CCTV</b>                   |
| 2 | <b>Stolen card warning</b>                             | 407 - Stolen Card Attempt     | CardID, EventCode, DoorID, Time             | <b>HR, CCTV</b>                   |
| 3 | <b>Warning: Resigned employees still swiping cards</b> | 410 - Terminate Card Attempt  | CardID, Department, DoorID, EventCode, Time | <b>HR</b>                         |
| 4 | <b>Wrong door/wrong area card swipe warning</b>        | 405 - Wrong Door Attempt      | CardID, DoorID, EventCode, Time             | <b>HR, CCTV</b>                   |
| 5 | <b>Door hold warning/door not closed</b>               | 903 - Door Held Open          | DoorID, EventCode, Time                     | <b>CCTV</b>                       |
| 6 | <b>The event automatically opens</b>                   | 437 - Auto unlocked           | DoorID, EventCode, ScheduleID, Time         | <b>BHS (luggage area related)</b> |
| 7 | <b>Automatic door lock event</b>                       | 438 - Auto-locked             | DoorID, EventCode, ScheduleID, Time         | <b>BHS (luggage area related)</b> |
| 8 | <b>Forced door event</b>                               | 900 - Door Forced Open        | DoorID, EventCode, Time                     | <b>CCTV</b>                       |

## 7.2 Use case by business (recommended)

### 7.2.1 Lost / Stolen Card Attempt

|                       |                                                                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>    | UC-406-407                                                                                                                                      |
| <b>Use Case Name</b>  | Lost / Stolen Card Attempt                                                                                                                      |
| <b>Description</b>    | When a lost or stolen card is swiped at any door, the system records and activates the area camera.                                             |
| <b>Actor</b>          | ACS, IB, CCTV, HR                                                                                                                               |
| <b>Trigger</b>        | The access event records EventCode=406 or 407 when swiping the card.                                                                            |
| <b>Pre-Conditions</b> | <ul style="list-style-type: none"> <li>- Card marked "Lost" or "Stolen" in the ACS database</li> <li>- IB, CCTV, HR working normally</li> </ul> |

|                         |                                                                                                                                                                                                             |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Post-Conditions</b>  | <ul style="list-style-type: none"> <li>- Event is logged EventCode=406 or EventCode=407</li> <li>- Send event to IB- IB sends to CCTV to turn on liveview camera on screen, send a warning to HR</li> </ul> |
| <b>Business Rules</b>   | Send minimum data to IB: CardID, DoorID, EventCode=406 or 407, UTC_Time.                                                                                                                                    |
| <b>Exceptional Flow</b> | If CCTV, IB, and HR lose connection, ACS only logs without sending a verification alert.                                                                                                                    |

## 7.2.2 Terminate Card Attempt (card assigned to employees leaving the job)

|                         |                                                                                                                            |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>      | UC-410                                                                                                                     |
| <b>Use Case Name</b>    | Terminate Card Attempt (card assigned to employees leaving the job)                                                        |
| <b>Description</b>      | Detected card attached to resigned employee used to swipe at the control gate.                                             |
| <b>Actor</b>            | ACS, IB, HR                                                                                                                |
| <b>Trigger</b>          | The access event records EventCode=410 when the card is swiped.                                                            |
| <b>Pre-Conditions</b>   | <ul style="list-style-type: none"> <li>- Employee data between ACS and HR is synchronized with each other</li> </ul>       |
| <b>Post-Conditions</b>  | <ul style="list-style-type: none"> <li>- ACS displays warnings, and event logs are sent to HR for verification.</li> </ul> |
| <b>Business Rules</b>   | Minimum data sent: CardID, Department, DoorID, EventCode, UTC_Time.                                                        |
| <b>Exceptional Flow</b> | If the HR, IB server does not respond, ACS only logs.                                                                      |

## 7.2.3 Wrong Door Attempt

|                       |                                                                                                                                                                           |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>    | UC-405                                                                                                                                                                    |
| <b>Use Case Name</b>  | Wrong Door Attempt                                                                                                                                                        |
| <b>Description</b>    | User swipes card at door without authorization; system displays warning and activates area video retrieval.                                                               |
| <b>Actor</b>          | ACS, IB, CCTV, HR                                                                                                                                                         |
| <b>Trigger</b>        | Access denied event with EventCode=405.                                                                                                                                   |
| <b>Pre-Conditions</b> | <ul style="list-style-type: none"> <li>- ACS has a credential record (cardID/userID) but does not have access to DoorID</li> <li>- IB, CCTV, HR work normally.</li> </ul> |

|                         |                                                                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Post-Conditions</b>  | <ul style="list-style-type: none"> <li>- Event is logged EventCode=405</li> <li>- Send event to IB, IB sends to CCTV to turn on the liveview camera on the screen</li> </ul> |
| <b>Business Rules</b>   | Send minimum data to IB: CardID, DoorID, EventCode=405, UTC_Time.                                                                                                            |
| <b>Exceptional Flow</b> | If CCTV, IB, and HR lose connection, ACS only logs without sending a verification alert.                                                                                     |

## 7.2.4 Door Held Open (Door opened over time)

|                         |                                                                                                                                                                                                                       |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>      | UC-903                                                                                                                                                                                                                |
| <b>Use Case Name</b>    | Door Held Open (Door held open over time)                                                                                                                                                                             |
| <b>Description</b>      | <ul style="list-style-type: none"> <li>- Door detection system for doors opened beyond the specified time</li> <li>- Activate the camera, push a warning to CCTV via IB to handle operations and security.</li> </ul> |
| <b>Actor</b>            | ACS, IB, CCTV                                                                                                                                                                                                         |
| <b>Trigger</b>          | Door open sensor exceeds timeout threshold.                                                                                                                                                                           |
| <b>Pre-Conditions</b>   | <ul style="list-style-type: none"> <li>- ACS and door sensor are working</li> <li>- IB, CCTV connection normal</li> </ul>                                                                                             |
| <b>Post-Conditions</b>  | Event is logged EventCode=903.<br>Warning displayed on ACS, siren/warning light at activated area, send event to IB, IB sends to CCTV, turn on liveview camera on screen.                                             |
| <b>Business Rules</b>   | Send minimum data to IB: DoorID, EventCode=903, UTC_Time.                                                                                                                                                             |
| <b>Exceptional Flow</b> | If CCTV, IB loses connection, a warning is displayed on ACS only, and on-site inspection is recommended.                                                                                                              |

## 7.2.5 Auto Unlock / Lock Door (door opens or locks automatically)

|                      |                                                                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>   | UC-437-438                                                                                                                                 |
| <b>Use Case Name</b> | Auto Unlock / Lock Door (door opens or locks automatically)                                                                                |
| <b>Description</b>   | The system automatically opens or locks the door according to the configured schedule, synchronizing events with the baggage system (BHS). |
| <b>Actor</b>         | ACS, IB, BHS                                                                                                                               |

|                         |                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------|
| <b>Trigger</b>          | The access event records EventCode=437 or 438 (Auto unlocked / Auto locked).                      |
| <b>Pre-Conditions</b>   | - Door operation schedule is set in ACS<br>- BHS connection is normal.                            |
| <b>Post-Conditions</b>  | Events are logged, door status is synchronized with BHS, and displayed on the operator interface. |
| <b>Business Rules</b>   | Send data: DoorID, EventCode, UTC_Time.                                                           |
| <b>Exceptional Flow</b> | If IB, BHS do not respond, ACS only saves the event log.                                          |

## 7.2.6 Door Forced Open

|                         |                                                                                                                                                                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>      | UC-900                                                                                                                                                                                                                                             |
| <b>Use Case Name</b>    | Door Forced Open                                                                                                                                                                                                                                   |
| <b>Description</b>      | - The system detects the door open sensor while the ACS does not receive a valid unlock signal - the system reports the “Door Forced Open” event<br>- Activates the camera, pushes the warning to CCTV via IB for operation and security handling. |
| <b>Actor</b>            | ACS, IB, CCTV                                                                                                                                                                                                                                      |
| <b>Trigger</b>          | Door contact changes to the OPEN state, but no corresponding Unlock/Grant event existed in the short period before.                                                                                                                                |
| <b>Pre-Conditions</b>   | - ACS, Door controller, and door contact sensor are online.<br>- IB and CCTV are connected normally.                                                                                                                                               |
| <b>Post-Conditions</b>  | - Event is logged EventCode=900<br>- Warning displayed on ACS, siren/warning light in the activated area, send event to IB, IB sends to CCTV to turn on liveview camera on screen                                                                  |
| <b>Business Rules</b>   | - Send minimum data to IB: DoorID, EventCode=900, UTC_Time                                                                                                                                                                                         |
| <b>Exceptional Flow</b> | - If CCTV, IB loses connection, a warning will only be displayed on ACS, and on-site inspection is recommended                                                                                                                                     |

## 8. Non-functional requirements

### 8.1. Performance

- Response time to alarms from ACS:  $\leq 200$  ms.
- Ability to process and transmit messages to integrated systems via IB:  $\geq 500$  msg/s.
- P95 message transmission delay:  $< 50$  ms.

## 8.2. Security

- All API/SDK communication between ACS and IB is encrypted with at least TLS 1.2.
- Apply RBAC (Role-Based Access Control) to users accessing ACS: Operator, Supervisor, Admin.
- The alarm payload between the ACS and IB is authenticated to ensure integrity.
- The system complies with ISO/IEC 27001 and ACIS-TSA security standards (where applicable).

## 8.3. Compliance with regulations

- Protection of personal data in compliance with Decree 13/2023/ND-CP.
- Data Governance Policy: data partitioning.
- Audit-trail: records User ID, time, action (Reset, Export report,...).
- Support for data deletion requests under GDPR/PDPA (right to be forgotten).

## V. FUNCTIONAL REQUIREMENTS

### V.1 Overview

ACS provides standard APIs for data sharing and control with IB. IB is responsible for receiving, normalizing, and distributing data to other systems.

### V.2 Access Control Data Services

#### V.2.1 General

The following general conventions apply to the Pro-Watch Web API.

##### URL Structure

“http://<ServerName>:<Port>/prowatch/api/<Method>”

##### Authentication

- API methods require authentication.
- Basic Authentication (username/password).
- Token Authentication (used for sessions after login).
- Authentication information must be included in the HTTP Header.

##### Request / Response Format

- Content-Type: “application/json”
- Request Body: JSON (for POST/PUT methods).
- Response Body: JSON (return value or status messages).

##### HTTP Methods

- GET - Query data (example: GetAlarms, GetDoors).

- POST - execute commands or create new object (example: DoorOpen, DoorLock, AddUser).
- PUT - Update existing data.

## HTTP Response Codes

| HTTP code                   | Description                                                               |
|-----------------------------|---------------------------------------------------------------------------|
| 200 - OK                    | Request successful, return data                                           |
| 204 – No Content            | Request successful, but no content returned                               |
| 400 – Bad Request           | The request has an invalid format or is missing required parameters.      |
| 401 – Unauthorized          | Invalid or missing authentication information (token, login).             |
| 403 – Forbidden             | No permission to call the API or insufficient privileges on the resource. |
| 404 – Not Found             | Object not found                                                          |
| 500 – Internal Server Error | Server error.                                                             |

### Example:

```
{
  "errorCode": "401",
  "errorMessage": "Unauthorized - Invalid API Key",
  "timestamp": "2025-08-20T09:00:00Z"
}
```

## V.2.2 Common Data Elements and Attributes

### • HardwareID

Hardware IDs follow the Pro-Watch Naming Convention. Plain English names are used, with a series of numbers to identify the piece of hardware:



**Site Name:** Name of site / system (Example: PW6000).

**Channel Panel:** Main channel board or control panel

**Subpanel:** Sub channel board or sub module.

**Logical Device:** Logical device (Example: door, reader, input, output).

### • EventCode



## EventCode List (PW-5000, PW-7000)

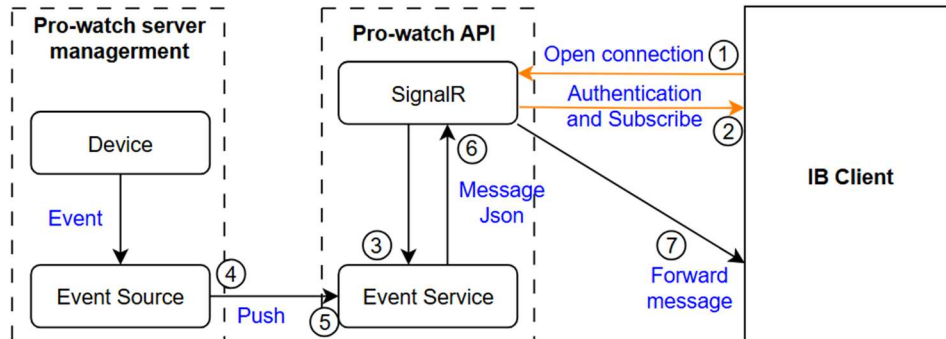
| EventCode | Description                                                        |
|-----------|--------------------------------------------------------------------|
| 400       | Unknown Card - Card does not exist in the system                   |
| 401       | Void Card - Invalid card, permanently disabled.                    |
| 402       | Expired Card Attempt - Card has expired.                           |
| 406       | Lost Card Attempt                                                  |
| 407       | Stolen Card Attempt                                                |
| 408       | Unaccounted for Card Attempt - Card not assigned to any user       |
| 409       | Deactivated Card Attempt                                           |
| 410       | Terminated Card Attempt                                            |
| 423       | Locked Attempt                                                     |
| 437       | Auto unlocked                                                      |
| 438       | Auto locked                                                        |
| 405       | Valid card at an unauthorized reader - Card used on the wrong door |
| 903       | Held door                                                          |

### Timestamp

- Definition: The timestamp when the event is recorded in the Pro-Watch system.
- Data field: EventDate, SystemDate
- Format: YYYY-MM-DDTHH:MM:SS
- Example: 2025-08-21T14:30:45
- **With:**
  - YYYY = Year
  - MM = Month
  - DD = Day
  - T = Separator between date and time
  - HH = Hour in UTC (00-23)
  - MM = Minute
  - SS = Second

## V.2.3 Operational Access Data (Event Service)

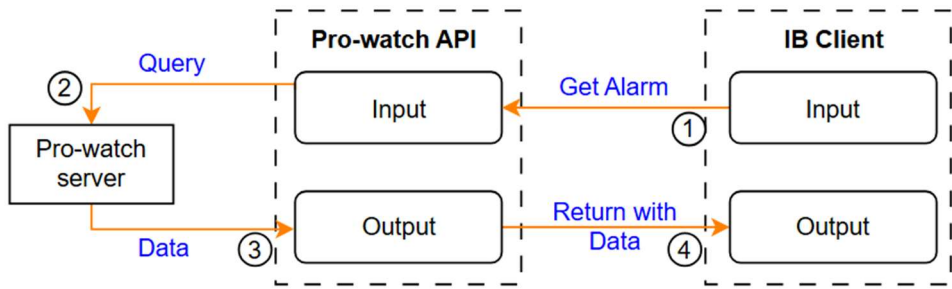
### A. GeteventRealTime (SignalR)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                                        |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------|
| <b>Interface name:</b>                       | SignalR – Subscribe Access Events                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |                                                        |
| <b>Description</b>                           | <p>Establish a real-time connection channel between the Pro-Watch Access Control System (ACS) and the Integration Broker (IB) via the SignalR Event Service.</p> <ul style="list-style-type: none"> <li>Purpose: IB receives all access events recorded in the Pro-Watch EV_LOG (e.g., Access Granted, Access Denied, Door Forced, Door Held, Door Status Change, Alarm, Fire/Emergency, etc.).</li> <li>IB acts as an integration middleware, receiving events from ACS and then routing and distributing them to other systems, such as CCTV (pop-up, recording), HR, Fire Alarm, Visitor Management, etc.</li> </ul>                    |             |                                                        |
| <b>Data Flow</b>                             |  <p>The diagram illustrates the data flow between the Pro-watch server management, Pro-watch API, and IB Client.          1. Open connection (1) from IB Client to SignalR.         2. Authentication and Subscribe (2) from IB Client to SignalR.         3. Message Json (3) from SignalR to Event Service.         4. Event Source (4) from Device to Event Source.         5. Push (5) from Event Source to Event Service.         6. Message Json (6) from Event Service to SignalR.         7. Forward message (7) from SignalR to IB Client.</p> |             |                                                        |
| <b>Input Parameters</b>                      | <b>Parameter Name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Type</b> | <b>Comment</b>                                         |
|                                              | AccessToken                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | String      | Authentication Token when connecting to SignalR        |
|                                              | SubscribedEvents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Array       | Event Types List (leave empty to receive all events)   |
|                                              | ClientID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | String      | IB Client Identifier to manage the session and logging |
|                                              | MaxResults                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Integer     | Number of Records Returned                             |
| The <b>AccessToken</b> parameter is required |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                                        |

| Output Parameters | Parameter Name                                                                                                                                                                                                                                                                                                                                                                   | Type     | Comment                                                                                 |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------|
|                   | EventId                                                                                                                                                                                                                                                                                                                                                                          | String   | Unique Event ID in EV_LOG                                                               |
|                   | EventType                                                                                                                                                                                                                                                                                                                                                                        | String   | Event Type (e.g., AccessGranted, AccessDenied, DoorForced, DoorHeld, Alarm, Fire, etc.) |
|                   | EventCode                                                                                                                                                                                                                                                                                                                                                                        | String   | Pro-Watch Internal Event Code                                                           |
|                   | EventDate                                                                                                                                                                                                                                                                                                                                                                        | DateTime | Event Timestamp                                                                         |
|                   | DoorId                                                                                                                                                                                                                                                                                                                                                                           | String   | Door / Reader ID                                                                        |
|                   | UserId                                                                                                                                                                                                                                                                                                                                                                           | String   | User ID                                                                                 |
|                   | BadgId                                                                                                                                                                                                                                                                                                                                                                           | String   | Card Number                                                                             |
|                   | DeviceId                                                                                                                                                                                                                                                                                                                                                                         | String   | Reader / Controller Generating the Event                                                |
|                   | Location                                                                                                                                                                                                                                                                                                                                                                         | String   | Device / Door Location                                                                  |
|                   | Priority                                                                                                                                                                                                                                                                                                                                                                         | Integer  | Event Priority Level                                                                    |
|                   | Message                                                                                                                                                                                                                                                                                                                                                                          | String   | Additional or accompanying message.                                                     |
|                   | The parameters EventId, EventType, EventCode, and EventDate are required.                                                                                                                                                                                                                                                                                                        |          |                                                                                         |
| Result Body       | Events are sent from Pro-Watch to IB as JSON objects via the SignalR channel.                                                                                                                                                                                                                                                                                                    |          |                                                                                         |
| Input sample      | <pre> javascript  const connection = new signalR.HubConnectionBuilder()     .withUrl("https://&lt;prowatch-server&gt;/signalr/eventhub?access_token=&lt;token&gt;")     .build();  connection.on("onProwatchEvent", function(eventData) {     console.log("Received Event:", eventData); });  connection.start().catch(err =&gt; console.error(err));                     </pre> |          |                                                                                         |

|                      |                                                                                                                                                                                                                                                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Output sample</b> | <pre> json {   "EventId": "EV123456",   "EventType": "AccessDenied",   "EventCode": "CARD_INVALID",   "EventDate": "2025-08-20T09:15:00Z",   "DoorId": "DOOR-01",   "UserId": "USR-1001",   "BadgeId": "BADGE-778899",   "DeviceId": "READER-DOOR-01",   "Location": "Main Entrance",   "Priority": 2,   "Message": "Access denied - invalid badge" } </pre> |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## B. GetAllDoorAlarm

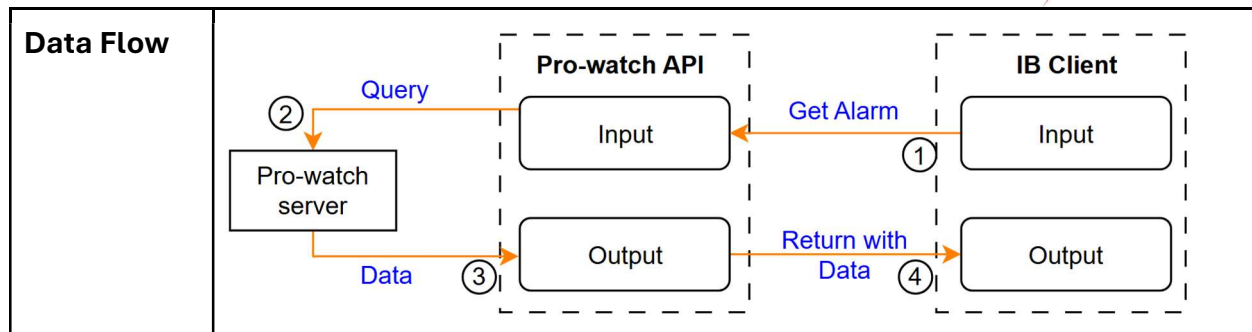
| Interface name:  | GET http://{server}:{port}/pwapi/alarms<br>Headers:<br>X-PW-WRKST: {WorkstationName}<br>Accept: application/json                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                   |  |                |      |         |            |        |                                                   |               |        |                      |      |        |                              |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|--|----------------|------|---------|------------|--------|---------------------------------------------------|---------------|--------|----------------------|------|--------|------------------------------|
| Description      | This function is used to retrieve the complete list of Alarm Events in the Pro-Watch ACS.<br>The API returns a list of events (event logs) recorded in EV_LOG, including events related to doors, cards, hardware, etc.                                                                                                                                                                                                                                                                                                            |                                                   |  |                |      |         |            |        |                                                   |               |        |                      |      |        |                              |
| Data Flow        |  <pre>graph LR     subgraph Pro-watch_API [Pro-watch API]         direction TB         PWS[Pro-watch server]         PWA[Input]         PWO[Output]         PWS -- "Query (2)" --&gt; PWA         PWA -- "Data (3)" --&gt; PWO     end     subgraph IB_Client [IB Client]         direction TB         ICI[Input]         ICO[Output]         ICI -- "Get Alarm (1)" --&gt; PWA         PWO -- "Return with Data (4)" --&gt; ICO     end</pre> |                                                   |  |                |      |         |            |        |                                                   |               |        |                      |      |        |                              |
| Input Parameters | <table><tr><th>Parameter Name</th><th>Type</th><th>Comment</th></tr><tr><td>X-PW-WRKST</td><td>String</td><td>Registered Workstation Name in Pro-Watch (header)</td></tr><tr><td>Authorization</td><td>String</td><td>Authentication Token</td></tr><tr><td>Host</td><td>String</td><td>Pro-Watch REST Server + Port</td></tr></table>                                                                                                                                                                                             |                                                   |  | Parameter Name | Type | Comment | X-PW-WRKST | String | Registered Workstation Name in Pro-Watch (header) | Authorization | String | Authentication Token | Host | String | Pro-Watch REST Server + Port |
| Parameter Name   | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Comment                                           |  |                |      |         |            |        |                                                   |               |        |                      |      |        |                              |
| X-PW-WRKST       | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Registered Workstation Name in Pro-Watch (header) |  |                |      |         |            |        |                                                   |               |        |                      |      |        |                              |
| Authorization    | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Authentication Token                              |  |                |      |         |            |        |                                                   |               |        |                      |      |        |                              |
| Host             | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Pro-Watch REST Server + Port                      |  |                |      |         |            |        |                                                   |               |        |                      |      |        |                              |

|                          |                                                                                                                                                                                                 |             |                                                                    |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------|
|                          | filter                                                                                                                                                                                          | OData       | OData Filter for results (Example: EventDate: 2025-08-21T09:00:00) |
|                          | All parameters Host, X-PW-WRKST, and Authorization are required.                                                                                                                                |             |                                                                    |
| <b>Output Parameters</b> | <b>Parameter Name</b>                                                                                                                                                                           | <b>Type</b> | <b>Comment</b>                                                     |
|                          | EventID                                                                                                                                                                                         | String      | Event GUID                                                         |
|                          | EventDate                                                                                                                                                                                       | DateTime    | Event Occurrence Timestamp                                         |
|                          | SystemDate                                                                                                                                                                                      | DateTime    | System Recorded Timestamp                                          |
|                          | EventDesc                                                                                                                                                                                       | String      | Event Description                                                  |
|                          | EventTypeID                                                                                                                                                                                     | GUID        | Event Type ID                                                      |
|                          | EventTypeDesc                                                                                                                                                                                   | String      | Event Type Name (example: Forced Door).                            |
|                          | EventCode                                                                                                                                                                                       | Int         | Event Code (example: 903 = Held Door)                              |
|                          | Priority                                                                                                                                                                                        | Int         | Alarm Priority Level                                               |
|                          | IsAlarm                                                                                                                                                                                         | Boolean     | Alarm Indicator                                                    |
|                          | LogicalDevice                                                                                                                                                                                   | Object      | Logical Device Information (ID, Description, Location, etc.)       |
|                          | HardwareID                                                                                                                                                                                      | String      | Hardware Identification (Panel, Reader, etc.)                      |
|                          | HardwareType                                                                                                                                                                                    | Int         | Hardware Type                                                      |
|                          | Badge                                                                                                                                                                                           | Object      | Cardholder Information                                             |
|                          | Card                                                                                                                                                                                            | Object      | Card Information                                                   |
|                          | Message                                                                                                                                                                                         | String      | Additional or accompanying message.                                |
| <b>Result Body</b>       | Response Status: HTTP 200 OK<br>Returned Data: JSON array                                                                                                                                       |             |                                                                    |
| <b>Input sample</b>      | GET /pwapi/alarms?\$top=50&\$orderby=EventDate desc HTTP/1.1<br>Host: {server}:{port}<br>X-PW-WRKST: FIRE-IB-WKS<br>Authorization: Basic ZGVtbzpwYXNzd29yZA==<br>Content-Type: application/json |             |                                                                    |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Output sample</b> | <pre> json  {   "EventID": "EventIDGUID",   "EventDate": "2025-08-20T13:45:00",   "SystemDate": "2025-08-20T13:45:01",   "EventDesc": "Door Locked",   "EventTypeID": "EventTypeGUID",   "EventTypeDesc": "Access Control",   "EventCode": 22,   "Priority": 1,   "IsAlarm": true,   "LogicalDevice": {     "LogDevID": "LogDevGUID",     "Description": "Reader 01",     "AltDescription": "North Gate",     "Location": "Terminal 1"   },   "HardwareID": "PW5000::Reader01",   "HardwareType": 9,   "Badge": {"BadgeID": "0x0055AA33BB66CC77",...},   "Card": {"CardNumber": "" },   "Message": "Door command executed successfully" } </pre> |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## C. GetForcedDoorEvents

|                        |                                                                                                                                                                                                            |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Interface name:</b> | <p>GET</p> <p>http://{server}:{port}/pwapi/alarms?eventdate={StartTime}&amp;eventdate={EndTime}&amp;eventCode=423</p> <p>Headers:</p> <p>X-PW-WRKST: {WorkstationName}</p> <p>Accept: application/json</p> |
| <b>Description</b>     | <p>This API is used to retrieve the list of alarms from the Pro-Watch system. Clients can filter by EventTypeDesc or EventCode to detect Door Forced Open events (unauthorized door opening).</p>          |



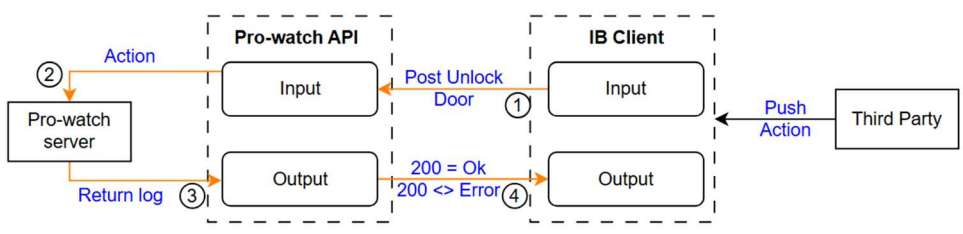
| Input Parameters  | Parameter Name                     | Type     | Comment                                                      |
|-------------------|------------------------------------|----------|--------------------------------------------------------------|
|                   | X-PW-WRKST                         | String   | Registered Workstation Name in Pro-Watch (header)            |
|                   | Authorization                      | String   | Authentication Token                                         |
|                   | Host                               | String   | Pro-Watch REST Server + Port                                 |
|                   | All input parameters are required. |          |                                                              |
| Output Parameters | Parameter Name                     | Type     | Comment                                                      |
|                   | EventID                            | String   | Event GUID                                                   |
|                   | EventDate                          | DateTime | Event Occurrence Timestamp                                   |
|                   | SystemDate                         | DateTime | System Recorded Timestamp                                    |
|                   | EventDesc                          | String   | Event Description                                            |
|                   | EventTypeID                        | GUID     | Event Type ID                                                |
|                   | EventTypeDesc                      | String   | Event Type Name (example: Force Door).                       |
|                   | EventCode                          | Int      | Event Code (example: 423 = Force Door)                       |
|                   | Priority                           | Int      | Alarm Priority Level                                         |
|                   | IsAlarm                            | Boolean  | Alarm Indicator                                              |
|                   | LogicalDevice                      | Object   | Logical Device Information (ID, Description, Location, etc.) |
|                   | HardwareID                         | String   | Hardware Identification (Panel, Reader, etc.)                |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |                                     |  |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------------------------------|--|
|                      | HardwareType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Int    | Hardware Type                       |  |
|                      | Badge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Object | Cardholder Information              |  |
|                      | Card                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Object | Card Information                    |  |
|                      | Message                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | String | Additional or accompanying message. |  |
| <b>Result Body</b>   | <p>Response Status: HTTP 200 OK<br/>Returned Data: JSON array</p> <p>EventTypeDesc: "Forced Door" – Type of event for unauthorized door opening<br/>EventCode: 423 – Event code for Forced Door<br/>IsAlarm: true – Marked by the system as an alarm</p>                                                                                                                                                                                                                                                                                                                                                                            |        |                                     |  |
| <b>Input sample</b>  | <p>GET /pwapi/alarms HTTP/1.1<br/>Host: {server}:{port}<br/>X-PW-WRKST: MY-WRKSTATION<br/>Accept: application/json</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |                                     |  |
| <b>Output sample</b> | <pre>json  {   "EventID": "EventIDGUID",   "EventDate": "2025-08-20T13:45:00",   "SystemDate": "2025-08-20T13:45:01",   "EventDesc": "Input point in alarm",   "EventTypeID": "EventTypeGUID",   "EventTypeDesc": "Forced Door",   "EventCode": 900,   "Priority": 12,   "IsAlarm": true,   "LogicalDevice": {     "LogDevID": "LogDevGUID",     "Description": "Reader 01",     "AltDescription": "North Gate",     "Location": "Terminal 1"   },   "HardwareID": "PW5000::Reader01",   "HardwareType": 9,   "Badge": { "BadgeID": "0x0055AA33BB66CC77" },   "Card": { "CardNumber": "" },   "Message": "Door forced open" }</pre> |        |                                     |  |



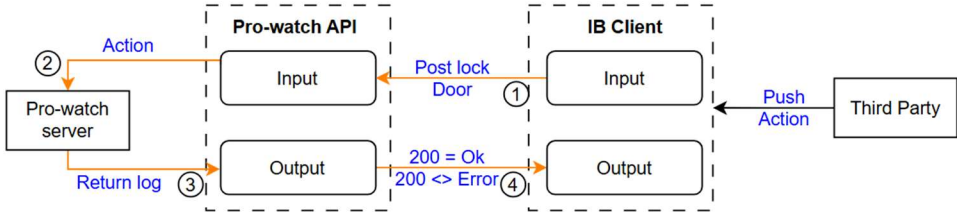
|  |   |
|--|---|
|  | } |
|--|---|

## D. Unlock Door Status

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                                                   |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------------------|
| <b>Interface name:</b>   | POST http://{server}:{port}/pwapi/logdevs/{logdev}/unlock                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |                                                   |
| <b>Description</b>       | <p>This function is used to perform an emergency door release in the Pro-Watch ACS system.</p> <p>Upon receiving a fire alarm signal, IB calls this API to send a door unlock command to the specified Logical Device (door).</p> <p>When the command is received, the system returns HTTP 200 OK.</p> <p>To verify whether the door has opened, an Alarm Event record is generated. Therefore, to confirm the command execution, it is necessary to call the GET /alarms API and filter by LogDevID and timestamp.</p>                                                                                                       |             |                                                   |
| <b>Data Flow</b>         |  <pre> graph LR     subgraph Pro-watch_API [Pro-watch API]         direction TB         PWS[Pro-watch server]         PWA[Pro-watch API]         PWO[Pro-watch API]     end     subgraph IB_Client [IB Client]         direction TB         IBW[IB Client]         IBOW[IB Client]     end     TP[Third Party]          TP -- "Push Action" --&gt; IBW     IBW -- "1 Post Unlock Door" --&gt; PWA     PWS -- "2 Action" --&gt; PWA     PWO -- "3 Return log" --&gt; PWS     PWO -- "4 200 = Ok, 200 &lt;&gt; Error" --&gt; IBOW     </pre> |             |                                                   |
| <b>Input Parameters</b>  | <b>Parameter Name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Type</b> | <b>Comment</b>                                    |
|                          | Host                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | String      | Pro-Watch REST Server + Port                      |
|                          | X-PW-WRKST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | String      | Registered Workstation Name in Pro-Watch (header) |
|                          | logdev                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | GUID        | Logical Device ID (the door to be unlocked)       |
|                          | All input parameters are required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                                   |
| <b>Output Parameters</b> | <p>Response Status: HTTP 200 OK (any status other than 200 indicates an error)</p> <p>Return a message: OK</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |                                                   |
| <b>Result Body</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                                                   |

|                      |                                                                                                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Input sample</b>  | POST /pwapi/logdevs/0x0071ABCD1234/unlock HTTP/1.1<br>Host: {server}:{port}<br>X-PW-WRKST: FIRE-IB-WKS<br>Authorization: Basic ZGVtbzpwYXNzd29yZA==<br>Content-Type: application/json |
| <b>Output sample</b> | HTTP/1.1 200 OK                                                                                                                                                                       |

## E. Lock Door Status

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                                                   |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------------------|
| <b>Interface name:</b>   | POST http://{server}:{port}/pwapi/logdevs/{logdev}/lock                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |                                                   |
| <b>Description</b>       | This function is used to lock a door after an emergency situation has ended. When this API is called, Pro-Watch receives the command and returns HTTP 200 OK, then executes the physical door lock. The door lock status is recorded in the Alarm Log.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |                                                   |
| <b>Data Flow</b>         |  <pre> graph LR     subgraph Pro-watch_API [Pro-watch API]         direction TB         PWS[Pro-watch server]         PWA[Pro-watch API]         PWO[Pro-watch API]         PWS -- "Action (2)" --&gt; PWA         PWA -- "Return log (3)" --&gt; PWO     end     subgraph IB_Client [IB Client]         direction TB         IWI[IB Client]         IWO[IB Client]     end     subgraph Third_Party [Third Party]         TP[Third Party]     end     TP -- "Push Action" --&gt; IWI     IWI -- "Post lock Door (1)" --&gt; PWA     PWO -- "200 = Ok (4)" --&gt; IWO     PWO -- "200 &lt;=&gt; Error (4)" --&gt; IWO </pre> |             |                                                   |
| <b>Input Parameters</b>  | <b>Parameter Name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Type</b> | <b>Comment</b>                                    |
|                          | Host                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | String      | Pro-Watch REST Server + Port                      |
|                          | X-PW-WRKST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | String      | Registered Workstation Name in Pro-Watch (header) |
|                          | logdev                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | GUID        | Logical Device ID (the door to be unlocked)       |
|                          | All input parameters are required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |                                                   |
| <b>Output Parameters</b> | Response Status: HTTP 200 OK (any status other than 200 indicates an error)<br>Return a message: OK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |                                                   |
| <b>Result Body</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                                                   |

|                      |                                                                                                                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Input sample</b>  | POST /pwapi/logdevs/0x0071ABCD1234/lock HTTP/1.1<br>Host: {server}:{port}<br>X-PW-WRKST: FIRE-IB-WKS<br>Authorization: Basic ZGVtbzpwYXNzd29yZA==<br>Content-Type: application/json |
| <b>Output sample</b> | HTTP/1.1 200 OK                                                                                                                                                                     |

## VI. DOCUMENTS

Long Thanh International Airport Passenger Terminal Construction Project

Link API and HSDK document: [Interface Prowatch](#)



## VII. ICD SUB-CONTRACTOR SIGN OFF SHEET